

Class 29 fractal structure

- The 2D system analysis (Gradescope) and weekly project update (canvas) are due this week.
- The 2D system analysis is an individual assignment with **no collaboration**. You may consult with course staff (individually) via office hours or by posting on Ed.

Big picture We are studying the geometric structures associated with chaotic dynamics. We saw this kind of structure begin to appear in the logistic orbit diagram. As a first introduction to that geometry, we will study the set of surviving points in the $3x$ tent map.

Activity

Teams 3 and 4: Post photos of your work to the course Google Drive today. Include words, labels, and other short notes that might make those solutions useful to you or your classmates. Find the link in Canvas.

Questions

1. (An interesting geometric structure. Example 4.9, Alligood et al) Consider the tent map $x_{n+1} = T_3(x_n)$, defined as

$$x_{n+1} = \begin{cases} 3x_n, & x_n \leq \frac{1}{2} \\ 3(1 - x_n), & \frac{1}{2} \leq x_n. \end{cases}$$

This is a slope-3 tent map. Consider it to be defined on the whole real line. Let C be the set of points on the real line whose orbits do not diverge to $-\infty$. This is the set of initial conditions where points in the orbit never become negative for T_3 .

The set C has an interesting (fractal) structure.

- (a) Sketch the map.
- (b) Initial conditions outside of $[0, 1]$ all have orbits that eventually diverge.
Convince yourselves that the interval $(1/3, 2/3)$ are the only points that leave the interval $[0, 1]$ under a single iteration of T_3 (note that $1/3$ and $2/3$ both stay in the interval).
Sketch the two line segments that are still in consideration for potentially being in C . Call this set C_1 .
- (c) What points will leave $[0, 1]$ under two iterations of T_3 ? Again sketch the line segments that are still in consideration for staying in the interval. Call this set C_2 . Is the set of points closed or open? (i.e. do the intervals contain their endpoints or not?)
- (d) Can you generalize this to sketch C_3 (three iterations)? What do you think will happen with k iterations?
- (e) What points seem to be in C ?

Answer:

a: It looks like a tent (stretching factor of 3).

b: Fixed points are 0 and $3(1 - x) = x$ so $3 - 3x = x$ so $x = 3/4$.

Mapping to zero: 0 and 1 map to zero.

Mapping to 0 or 1: $1/3$ maps to 1 and so does $2/3$.

Mapping to 0, 1, $1/3$, $2/3$: $1/9$ and $2/9$ map to $1/3$ and $2/3$. In addition, $7/9$ and $8/9$ do too.

c: the stuff that leaves is the stuff that maps to above 1. $f(1/3) = 1$ and $f(2/3) = 1$. For $1/3 < x < 2/3$, $f(x) > f(1/3)$, so that is the range that leaves. The sketch is two lines.

d: $[0, 1/9]$, $[2/9, 1/3]$, $[2/3, 7/9]$, $[8/9, 1]$ stay in, so those four segments should be the sketch.

e: Each segment in C_2 gets two little segments in C_3 . That will keep happening so lots of little segments in C_k .

f: The limit of this process: hard to describe!

2. The middle-thirds Cantor set, C , is formed by starting with $S_0 = [0, 1]$ and repeatedly removing the open middle third from every interval that remains. Let S_n be the set that remains after n steps, and let $C = S_\infty$.

(a) Sketch S_0 , S_1 , S_2 , and S_3 .

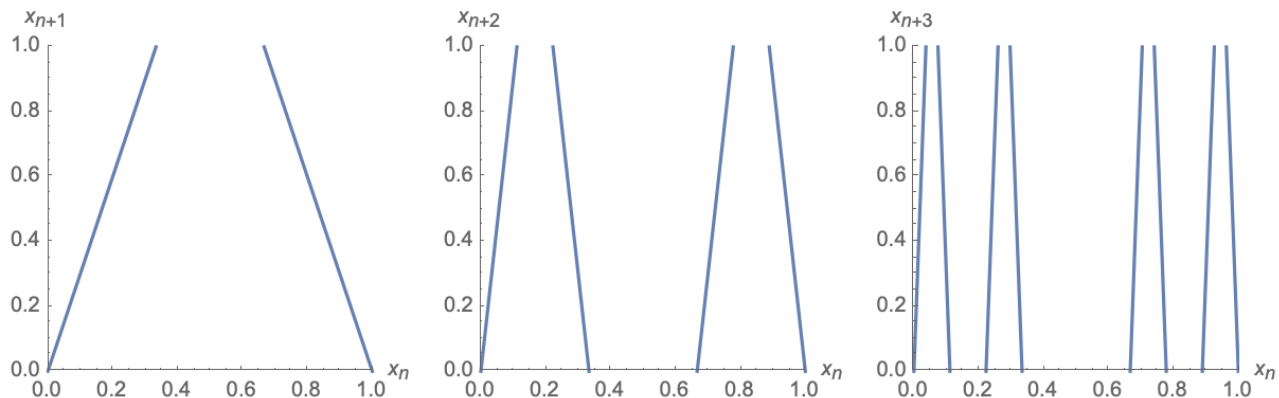
(b) Describe how S_{k+1} is built from scaled copies of S_k .

(c) How many intervals are in S_k ? What is the length of each interval?

(d) What is the total length of the set S_k ?

(e) What does this suggest about the total length of the Cantor set?

Repeated iterates of the 3x tent map



- The set of initial conditions that stay within the interval $[0, 1]$ under repeated iteration of the map is a Cantor set. This particular Cantor set is called the **middle thirds Cantor set**.
- A **Cantor set** is a limiting set, $C = S_\infty$, that arises from a repeated interval removal process.

3. We removed intervals of total length 1 to construct the Cantor set, and the set that remains has total length 0.

How does this set still contain many points?

- (a) At each step of the construction, a point in the Cantor set lies either in a left subinterval or a right subinterval. Use 0 to indicate left and 1 to indicate right. This gives each point an **itinerary** consisting of an infinite sequence of 0s and 1s.

What point is encoded by the itinerary 0111... (where the 1s continue to repeat)?

- (b) An infinite itinerary is associated with a nested sequence of intervals drawn from $S_1, S_2, \dots$. Sketch the first three nested intervals associated with 0111...

Each infinite itinerary determines a unique point in the Cantor set.

- (c) Points in the interval $[0, 1]$ can also be represented by infinite binary expansions $0.a_1a_2a_3\dots$. Does this suggest that the Cantor set has as many points as the interval $[0, 1]$?

- (d) Is every point in the Cantor set an endpoint of one of the intervals in some S_n ?

Consider the itinerary 010101..., with 01 repeating.

Answers:

- $1/3$.
- An interval of size $1/3$ with two subintervals each at the far right.
- Yes. For each point in $[0, 1]$, choose a binary expansion. Treat that binary expansion as an itinerary. This itinerary determines a unique point in the Cantor set. So we can assign a point in the Cantor set to each point in $[0, 1]$, which shows that the Cantor set has at least as many points as $[0, 1]$.
- no. endpoints are associated in ending in all 0s or all 1s after a certain n .

We'll call a set a **topological Cantor set** if it can be stretched and squished to look like a Cantor set. Such sets are totally disconnected and contain no isolated points. In the $3x$ tent map, repeated removal leaves a Cantor set. In the logistic orbit diagram, repeated period-doubling creates a distorted version of this same kind of structure.

Dimension of the Cantor set

- What is dimension? How do you identify the dimension of a set of points?
- If you were to zoom in on a piece of C , say the part in the rightmost line segment of S_2 , how would it look? Would it look different from all of C ?
- A set is **self-similar** when it contains smaller copies of itself at all scales.

4. A self-similar set is made of m scaled copies of itself, each reduced by a factor of r . Its **similarity dimension** d satisfies

$$m = r^d.$$

- (a) Using $r = 3$, find the similarity dimension of:

- the line segment $[0, 1]$,
- the middle-thirds Cantor set,

- the square $[0, 1] \times [0, 1]$.

Scaling by $r = 3$ replaces a square by a smaller square with sides $1/3$ the original.

- (b) Compare the dimension of the Cantor set to the dimensions of the line segment and the square.

answers:

Consider the line segment $[0, 1]$. Reconstruct it using scaled down copies. Choose a scaling of $r = 3$ so the copies are each of length $1/3$. $m = 3$ copies are needed to reconstruct the line segment, so $3 = 3^d$ and $d = 1$.

Consider the Cantor set. Reconstruct it using scaled down copies. Choose a scaling of $r = 3$ so the copies are each of length $1/3$. $m = 2$ copies are needed to reconstruct the set, so $2 = 3^d$ and $d = \ln 2 / \ln 3 \approx 0.63$.

Consider the square $[0, 1] \times [0, 1]$. Reconstruct it using scaled down copies. Choose a scaling of $r = 3$ so the copies are each of side length $1/3$. $m = 9$ copies are needed to reconstruct the square, so $9 = 3^d$ and $d = 2$.

Fractals

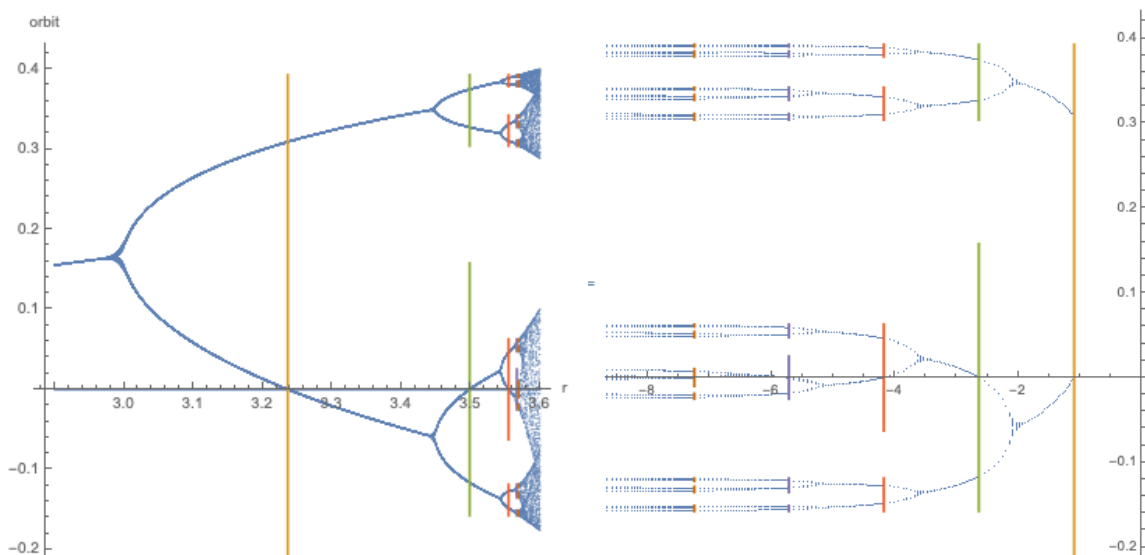
We'll use the term **fractal** to refer to sets that have:

- complicated structure at a wide range of length scales
- repetition of structure at different length scales (for example, **self-similarity**)
- a **fractal dimension** that is not an integer. *Definition from Alligood et al 2000.*

Fractals regularly arise in the structure of strange attractors.

5. Based on the diagram on the right below, how does the series of period doubling bifurcations in the logistic map lead to a Cantor-like, fractal structure (a "topological" Cantor set)?

In a topological Cantor set the gaps are various sizes and the set is not strictly self-similar.



On the left is the logistic map orbit diagram (for a logistic map centered about 0). On the right, I am showing r on a log scale based on the distance to r_∞ , the limiting r value associated with the period-doubling cascade.